

Peer-review-style critique of the fifteen-repository exoplanet portfolio

Scope and review standard

I reviewed the README and analysis script for all fifteen repositories, and the `index.html` report for each of the eleven planet-report repositories. I treated the portfolio as I would material submitted by a prospective graduate student or research-assistant applicant: not asking whether every repository rises to the standard of a refereed paper, but asking whether the author understands what the data actually constrain, distinguishes descriptive statistics from scientific inference, propagates uncertainty correctly, represents published work accurately, and writes code whose numerical output I could trust without first reverse-engineering it.

An important distinction should be made at the outset. The planet projects genuinely use **real observational products**, but in most cases the scripts operate on already-reduced spectra, phase-curve fit products, or literature/Zenodo tables, rather than carrying JWST/HST/Spitzer detector data through calibration, light-curve extraction, systematic detrending and astrophysical fitting. That is entirely legitimate portfolio work, but it makes these primarily **secondary quantitative analyses of published data products**, not independent reductions of archival observations. Several webpages blur that distinction by repeatedly stressing “real data” and “reproduces the detection”. The strongest versions of these projects are good exercises in scientific interpretation; they are not equivalent in depth to having independently reduced a JWST time series.

My standard for statements such as “ 12.9σ detection”, “molecular detection”, “rules out”, “direct evidence” and “systematic uncertainty” is correspondingly strict. Those words carry well-defined implications in exoplanet spectroscopy and should not be used for a statistic that is only a band-average difference, a range across pipelines, or an injected-signal recovery whose true parameters were used during the search.

Overall portfolio assessment

As a student portfolio, this is credible and materially better than a collection of plotting notebooks. As research evidence, however, it is currently uneven enough that I would not send the portfolio unchanged to an exoplanet-atmospheres PI. There is genuine competence here. The author is comfortable reading machine-readable science products, dealing with inverse-variance weighting, propagating elementary uncertainties, numerically inverting Planck functions, solving Kepler's equation, implementing the point-source/point-lens microlensing equation, handling phase curves, building ADI-style image combinations, and separating pedagogical simulations from observational datasets. The repositories are also unusually explicit about data provenance and limitations, and some deliberately report null results instead of forcing a detection. Those are good scientific instincts. For example, WASP-96b explicitly finds essentially no average limb asymmetry rather than manufacturing one; LHS 475b correctly recognises that a flat spectrum can constrain hydrogen-rich and methane atmospheres without proving a particular high- μ atmosphere; and HD 209458b draws attention to reduction-choice sensitivity, an issue that the source paper itself explicitly treats through a tree-structured reduction methodology. ¹

The **strongest evidence of scientific competence** is therefore not any one impressive-looking sigma value. It is the repeated ability to turn published products into a reproducible calculation, understand roughly what physical quantity is being measured, and in several cases articulate degeneracies and limitations. The clean, consistent presentation helps. The weakest evidence is the step from **calculation to inference**. Too often, a simple statistic is given a stronger scientific name than it deserves: a two-band contrast becomes a molecular detection significance; a maximum-minus-minimum across four reduction leaves becomes a “systematic uncertainty”; an arbitrary CO₂ band-minus-continuum difference becomes a direct test of a retrieval claim; an unweighted mean of point-estimate brightness temperatures becomes evidence for global heat redistribution; and an injected microlensing perturbation is “recovered” using the very true parameters that generated it. Those are not cosmetic issues. They are exactly the distinctions on which experienced astronomical data analysts judge whether someone understands inference rather than merely arithmetic.

For **graduate admissions**, I would regard the portfolio as evidence of strong initiative and useful preparation, but not yet as evidence that the applicant can independently conduct research-grade exoplanet inference. For an **astronomy research-assistant position**, I would want to see the major errors below corrected and at least one project pushed substantially deeper into covariance-aware fitting or end-to-end time-series reduction. For a **general data-science application**, the repositories show numerical literacy and readable Python, but the statistical calibration issues actually matter there too: calling a quantity “significance” when its likelihood assumptions have not been established is a transferable methodological weakness, not astronomy-specific pedantry.

Per-repository critique

WASP-107b. This is a reasonable descriptive exploration of a real JWST/NIRISS transmission spectrum: the inverse-variance weighted mean is implemented correctly under independent Gaussian errors, the plotting is clear, and the webpage sensibly separates the NIRISS wavelength range from the MIRI results discussed in the literature. But the analysis itself is shallow: the headline peak-to-trough amplitude is simply `max(depth)-min(depth)`, with no propagated uncertainty, covariance treatment, or protection against an extreme noisy bin, and statements about the 1.4-μm structure being “water-consistent” are qualitative rather than a molecular test. The repo should also update its broader methane narrative: later JWST/NIRSpec observations detected CH₄ at 4.2σ , albeit at an abundance depleted by roughly three orders of magnitude relative to equilibrium expectations, so “absence of methane” is no longer a safe global characterisation of WASP-107b. The current webpage is polished and readable, but “reproducing atmospheric features” overstates what is essentially plotting plus summary statistics. ²

WASP-96b. Among the planet reports, this is conceptually one of the better ones because the author asks a falsifiable question—whether morning and evening limbs differ—and accepts the null answer. The weighted difference calculation and quadrature of the quoted per-limb formal errors are correct *if* the two estimates are independent. The important caveat is that they come from the same observations/reduction and can share stellar, instrumental and fitting covariance; a weighted mean of per-bin differences is therefore not a complete limb-asymmetry hypothesis test. The source study itself treats limb asymmetry cautiously and emphasises the need for robust inference at low signal-to-noise. The repo’s peak-to-trough spectral amplitude is again an uncertainty-free extreme-value statistic. Code quality is clean enough for exploratory work, and the webpage is strong, but I would describe the result as a **first-order diagnostic consistent with no average asymmetry**, not a reproduction of a rigorous asymmetry analysis. ³

K2-18b. This is scientifically interesting but considerably overclaimed. The script forms an inverse-variance mean in a 4.1–4.6- μm box and subtracts an arbitrarily selected 3.6–3.9- μm “continuum”, obtaining roughly 0.2σ . That is a legitimate descriptive colour/band statistic; it is **not a direct test of the CO₂ detection claim**, because CO₂ inference comes from the joint spectral morphology, competing molecular opacities, instrument offsets/systematics and atmospheric retrieval. The problem is amplified by performing the significance calculation over thousands of native-resolution points as though all were independent. Spectral extraction and instrumental systematics can introduce substantial correlation, and current K2-18b work explicitly finds that MIRI reductions are susceptible to red noise and binning choices. Recent independent analyses still favour CO₂ in the full spectrum even while rejecting significant biosignature evidence, which directly demonstrates why “my band statistic is 0.2σ ” cannot be translated into “the CO₂ claim disappears”. The code is readable, and displaying binned data is good, but the scientific conclusion must be downgraded sharply. 4

HD 209458b. The central idea is excellent for a portfolio: use four legitimate branches of a published tree-structured MIRI reduction to show how choices in data processing alter the transmission spectrum. That is much closer to an authentic methodological question than simply plotting a famous spectrum, and the 2026 source paper itself was designed around incorporating uncertain reduction decisions into subsequent modelling. The numerical interpretation is where the repo goes too far. The per-wavelength **max-min range across four leaves is not a statistically calibrated “systematic uncertainty”**, and it is not independent of photon noise; the four reductions share the same photons and much of the same processing. Comparing the mean range with the mean formal error can illustrate sensitivity to analysis choices, but it is not a variance decomposition. The README wording “first transiting exoplanet ever found” should also be tightened to “first known exoplanet observed to transit” or equivalent: HD 209458b was initially identified through radial velocities before the transit detection. Overall, this is one of the stronger ideas in the portfolio, weakened mainly by nomenclature. 5

TRAPPIST-1e. This needs an urgent rewrite before anyone in the field sees it. The README says that “TRAPPIST-1 e has no published atmospheric spectrum of its own yet”, but the repository itself was created on **11 August 2026**, while an *Astronomical Journal* paper presenting the first observations of the JWST TRAPPIST-1 e/b programme was published online on 22 January 2026 and in the February 2026 issue; NASA was already publicly presenting a TRAPPIST-1e NIRSpec transmission spectrum in 2025. The current observations do not establish an atmosphere—the stellar contamination problem remains serious—but saying no spectrum has been published is factually stale at the date the repo was created. 6 The code then analyses **TRAPPIST-1b only**, despite presentation language referring to “siblings b and c”. Its Planck inversion is a reasonable pedagogical approximation, but it evaluates monochromatic blackbodies at central wavelengths instead of integrating the planet/star spectra through the MIRI bandpasses, and the “bare rock limits” assume zero Bond albedo and particular redistribution prescriptions without carrying those assumptions into the headline. More importantly, one measured brightness temperature being above a simplified zero-albedo equilibrium estimate does not by itself prove an absence of atmospheric heat transport. The repo’s basic calculation is defensible; its framing around planet e and its current-state-of-literature claim are not.

WASP-121b. The data source is excellent: the underlying 2025 *Nature Astronomy* analysis uses a complete NIRSpec/G395H phase curve, 349 spectral channels and 36 orbital-phase bins, with explicit posterior uncertainties and atmospheric retrievals. 7 The portfolio script, however, discards much of that statistical content. It takes simple means in hand-selected “dayside” and “nightside” phase windows and reports approximately 2737 K versus 1176 K, but it does not use the provided asymmetric brightness-temperature uncertainty information and supplies no uncertainty on the ~ 1560 K contrast. Those numbers are fine as a visual summary, not as an uncertainty-aware measurement. The source paper interprets dayside and nightside spectra through posterior distributions, pressure-temperature profiles and retrievals; the repo reduces that to two arithmetic means. The description of inefficient

heat redistribution is qualitatively plausible, but comparing an absolute temperature contrast with other planets without normalising for irradiation, photospheric pressure and spectral weighting is too casual. The webpage looks very good; the analysis is substantially shallower than the sophistication of the dataset makes it appear. ⁷

GJ 1214b. The Planck inversion is implemented cleanly, but the uncertainty treatment is unacceptable for the strength of the stated conclusion. The source paper reports average dayside and nightside brightness temperatures of 553 ± 9 K and 437 ± 19 K and derives a Bond albedo of 0.51 ± 0.06 , with the atmospheric conclusion informed by spectra, energy balance and 3-D models. ⁸ The repo instead converts the median fitted flux at each wavelength into a brightness temperature, ignores the supplied posterior errors, takes an unweighted mean of those temperatures, and—most concerningly—uses `max(flux, 1e-8)` before inversion. Thus a fitted nightside flux that is zero or negative because of noise is silently converted into a tiny *positive* flux and hence a finite brightness temperature. That is statistically wrong; non-detections should produce posterior constraints or upper limits, not clipped measurements. The reported 583 K / 449 K mean temperatures are also not directly the same estimator as the source paper's band-integrated values. The conclusion that the phase curve supports an atmosphere is consistent with the published science, but **this script does not independently establish it.**

HD 189733b. This repository contains one of the clearest examples of a descriptive statistic being mislabeled as scientific detection significance. The code measures the difference between a 4.3–4.6- μ m band and a 3.6–3.9- μ m comparison region and calls the result a 22.1σ CO₂ detection; it similarly labels a 2.6–3.0- μ m contrast as a 24.7σ “H₂O/H₂S” detection. That is **not how molecular detection significance is established**. The published retrieval/model-comparison analysis reports H₂O at 13.4σ , CO₂ at 11.2σ , CO at 5σ and H₂S at 4.5σ . ⁹ A broadband depression can contain contributions from several absorbers, the continuum is model-dependent, and wavelength errors/systematics need not be independent. Calling 24.7σ a detection of a blended “H₂O/H₂S” band is particularly problematic because it cannot tell which molecule generated the contrast. The code itself is neat and easy to inspect; the interpretation is scientifically indefensible in its present wording. This is a case where simply relabelling the numbers as **empirical band-contrast S/N** would remove most of the problem.

55 Cancr e. This is another repository that I would pull from a public application until corrected. Its CSV contains two values, $0.0083\% \pm 0.0014\%$ and $0.0154\% \pm 0.0023\%$, labelled by publication month, and the script treats those as two independent observational epochs from which it derives a 2.7σ brightness-temperature change. But the 83 ± 14 ppm value is the **mean of eight Spitzer occultations observed in 2012–2013** in Demory et al.'s variability study, obtained by fitting all eight eclipses; it is not one epoch. ¹⁰ The 154 ± 23 ppm quantity is a separate published Spitzer measurement/analysis used elsewhere in the literature, not simply a second independent epoch matched to the first by publication date. ¹¹ Therefore subtracting the two catalogue values and labelling the result an epoch-to-epoch 2.7σ variability detection is not statistically meaningful. Ironically, the real variability of 55 Cnc e is well established from the actual multi-epoch Spitzer analyses and has since been observed dramatically with JWST, where five NIRCcam eclipses vary from non-detections to roughly 119 ppm at 4.5 μ m. ¹² The project has chosen the right astrophysical phenomenon but the wrong inferential construction.

HAT-P-11b. There is a serious provenance error here. The README and webpage repeatedly state that the Allart et al. (2018) high-resolution helium data are **HARPS-N/GIANO-B** observations. They are not: Allart et al.'s spectrally resolved HAT-P-11b helium detection was made with **CARMENES**. Independent HST/WFC3 observations by Mansfield et al. detected the feature at about 4σ . ¹³ The published CARMENES result gives average helium absorption of $1.08 \pm 0.05\%$, with the two transits at approximately $0.82 \pm 0.09\%$ and $1.21 \pm 0.06\%$. ¹⁴ The repository instead advertises $0.827 \pm 0.064\%$ at 12.9σ as the combined “real” detection. Its script defines transit using an ad hoc `|phase| <`

0.02 window and treats the phase-folded points as statistically independent; that is not a reproduction of the published combined measurement. The physical interpretation—HAT-P-11b possesses an extended escaping helium atmosphere—is correct, but getting the instrument wrong in an observational spectroscopy portfolio is precisely the sort of mistake a domain expert notices immediately.

LHS 475b. This is one of the more scientifically defensible planet projects. The flat-line weighted fit is appropriate as a first test, and the qualitative conclusion matches the literature: primordial H-dominated and cloud-free pure-CH₄ atmospheres are excluded, while a high-altitude-cloud/high- μ atmosphere, tenuous atmosphere, or effectively airless planet remains possible. ¹⁵ The weaknesses are statistical terminology. The script reports reduced χ^2 , but the webpage sometimes labels the same values simply “ χ^2 ”; for the H₂-rich model, “ $\chi^2 = 206$ ” is therefore ambiguous/mislabeled. A reduced- $\chi^2 > 2$ heuristic is also not a principled model-rejection threshold—one should report the actual χ^2 , degrees of freedom and p-value or, better, the same model-comparison statistic used in the source analysis. The webpage says the models are “offset to the measured depth”, whereas the inspected script does not itself fit an offset; that needs to be explicitly documented as part of the upstream model product if it was already done. Still, unlike several other reports, the central scientific conclusion is restrained and substantially correct.

Transit photometry detection method. Pedagogically, this is a decent introduction to why box-like periodic dips can be found with BLS-type searches, but it should not say or imply that it recreates a survey-grade BLS pipeline. The injection is a very easy white-Gaussian-noise problem, and the search is given the **true transit duration in advance**, scanning period and phase but not duration. Operational BLS searches allow a range of transit fractions/durations; the NASA Exoplanet Archive's implementation, for example, explicitly specifies minimum and maximum fractions of the period spent in transit. ¹⁶ The reported 74 σ is an in-transit depth S/N under the known white-noise model, not a search significance corrected for the period/phase trials. The synthetic light curve also lacks realistic red noise, detrending, data gaps, stellar variability and limb-darkened ingress/egress. The README is strong as teaching material; the numerical validation is too easy to demonstrate robust detection competence.

Radial-velocity detection method. The underlying Keplerian physics is broadly sound, and writing one's own Kepler-equation solver and RV model is useful. But there is an outright factual error in the README: in the $M_p \ll M_*$ limit, the RV semi-amplitude scales approximately as $M_*^{-2/3}$, **not** $1/\sqrt{M_*}$. The “from scratch” claim is also too expansive because the periodogram and nonlinear optimisation rely on SciPy's `lombscargle` and `curve_fit`; that is perfectly sensible coding, but it is not a from-scratch Lomb–Scargle implementation. More seriously, the fit has no physical bounds: eccentricity is clipped internally when evaluating the model, while the optimiser's actual eccentricity parameter may wander to an unphysical value that is subsequently used in the minimum-mass calculation. There is no false-alarm probability for the periodogram, no fitted stellar/instrumental jitter, no zero-point offsets or trends, and the injected $K \sim 92 \text{ m s}^{-1}$ signal against $\sim 1.8 \text{ m s}^{-1}$ noise makes the recovery essentially a demonstration that a very high-S/N Keplerian can be fitted. Good teaching exercise; not a convincing validation of real RV inference.

Gravitational microlensing detection method. This is the weakest scientific simulation in the portfolio. The point-source/point-lens magnification formula is correct, and it is good that the README openly says it is not solving a full binary-lens equation. But the claimed planet-mass-ratio “recovery” is essentially circular. The perturbation is injected as an artificial Gaussian whose duration/amplitude are defined by hand-tuned $\sqrt{q}$ relations, and the script then algebraically inverts those same relations to recover q . Worse, the code uses the **true injected perturbation time and true mass ratio to define the perturbation masking/window used during fitting**, so information unavailable to a real observer

leaks directly into the recovery. The statement that planetary-perturbation *amplitude* generically scales as $\sqrt{q}$ is also too strong: caustic size/duration have useful mass-ratio scalings, but perturbation amplitude is strongly dependent on source trajectory, caustic topology, projected separation, finite-source effects and blending. A 0.03% mass-ratio recovery under this construction is therefore not evidence that the method can infer planetary mass ratios. This would embarrass the author in front of a microlensing specialist unless the validation language is radically rewritten.

Direct imaging detection method. The broad ADI workflow is recognisable and pedagogically useful: inject a rotating source into a sequence with quasi-static structure, construct a reference PSF, subtract, derotate and combine. That is the right conceptual sequence, and Marois et al.'s ADI method indeed exploits field rotation to distinguish astrophysical companions from quasi-static speckles.¹⁷ The simulation is nevertheless much too idealised to support the quoted 38.8σ or its “ 5σ contrast curve”. The speckle field is an artificial Gaussian texture rather than a correlated, evolving PSF residual; the significance uses only a small set of comparison apertures and ignores the small-sample correction that is important at small angular separation—ironically despite citing Mawet et al., whose paper shows that naïve Gaussian thresholds can produce order-of-magnitude errors in false-alarm calibration.¹⁸ Most importantly, the plotted contrast limit is not properly calibrated as planet/star contrast: aperture residual noise must be normalised by stellar flux measured consistently and corrected for algorithmic throughput/self-subtraction. As written, the curve mixes residual flux units with the injected peak-amplitude contrast. The image-processing intuition is good; the sensitivity curve is not scientifically trustworthy.

Errors and red flags, ranked

The ranking below is by how damaging I think each issue would be if noticed during an application review, rather than by how easy it is to fix.

1. **Critical — HAT-P-11b: incorrect observing instrument and misleading reproduction claim.** Allart et al. 2018 used CARMENES, not HARPS-N/GIANO-B; the repo's $0.827 \pm 0.064\%$ result also does not reproduce the published combined $1.08 \pm 0.05\%$ measurement. This is a direct data-provenance error in an observational astronomy portfolio.¹⁴
2. **Critical — 55 Cancri e: invalid construction of “two independent epochs”.** The 83 ± 14 ppm quantity is a mean obtained from all eight 2012–2013 Spitzer eclipses, yet the repository treats it as one epoch and contrasts it with another catalogue/literature value based on publication date. The derived 2.7σ variability statistic therefore has no defensible sampling interpretation.¹⁹
3. **Critical — microlensing: validation leaks the injected truth.** The true perturbation location and mass ratio help define the fitting/masking window, and the mass ratio is recovered by inverting the same hand-set scaling law used to generate the perturbation. The quoted precision is circular, not an independent recovery.
4. **Critical/high — HD 189733b: empirical band S/N presented as molecular detection sigma.** The repo's 22.1σ CO₂ and 24.7σ H₂O/H₂S figures exceed and do not correspond to the source paper's model-based molecular significances of 11.2σ for CO₂, 13.4σ for H₂O and 4.5σ for H₂S. A “H₂O/H₂S” broadband statistic cannot itself identify either molecule.²⁰
5. **High — TRAPPIST-1e: factually outdated on the day it was created.** The repository was created on 11 August 2026 but states that no atmospheric spectrum of e has yet been published, despite published 2026 JWST programme results and publicly released NIRSpec transmission-

spectrum material. It also advertises b and c context while the inspected analysis operates on b.

21

6. **High — direct imaging: the “5 σ contrast curve” is not calibrated in contrast units and does not include throughput or small-number statistics.** That makes the sensitivity curve unsuitable as a scientific detection-limit product. 22
7. **High — K2-18b: a two-window spectral colour is presented as a direct test of CO₂.** It is not equivalent to a molecular retrieval and it treats a large number of spectral points as independent. Current independent analyses can simultaneously dispute the MIRI biosignature claim and still favour CO₂, demonstrating the inadequacy of the boxcar test for the stated inference. 23
8. **High — GJ 1214b: clipping non-positive fitted nightside flux to 10⁻⁸ before Planck inversion.** This turns statistical non-detections/negative fluctuations into positive temperatures and then averages them without posterior uncertainties.
9. **High/moderate — WASP-121b: supplied uncertainty information is discarded.** A 1500-K-plus day/night contrast is reported without propagating the phase-resolved brightness-temperature posteriors, despite the source analysis explicitly working with posterior samples. 7
10. **Moderate — HD 209458b: max-min reduction spread is called systematic uncertainty.** It is a useful sensitivity metric but not a statistically calibrated uncertainty and certainly not independent of photon noise. The source paper's tree framework is considerably more sophisticated. 24
11. **Moderate — transit method: the recovery is given the true duration and a very benign white-noise light curve.** The quoted detection S/N is not a trial-corrected BLS significance, so claims of survey-pipeline equivalence should be removed. 25
12. **Moderate — RV method: incorrect stellar-mass scaling plus an unconstrained fit.** $K \propto M_{\star}^{-2/3}$ in the planet-mass-negligible limit, not $M_{\star}^{-1/2}$; the optimiser can also return parameters inconsistent with the clipping used inside the forward model.
13. **Moderate — LHS 475b: reduced χ^2 is sometimes labelled plain χ^2 , and model rejection is reduced to an arbitrary threshold rather than a probability/model comparison.** The underlying qualitative conclusion is nevertheless consistent with the published paper. 26
14. **Moderate/minor — WASP-96b: the limb-difference errors assume independent limb measurements and the peak-to-trough amplitude has no uncertainty.** This is acceptable as an exploratory statistic if labelled accordingly, but not a definitive test of terminator symmetry. 27
15. **Minor/moderate — WASP-107b: extreme-bin amplitude and qualitative feature identification are presented with more confidence than the script supports, and the methane discussion needs updating for the later NIRSpec detection.** The core spectrum and calculations are otherwise legitimate. 28

The pattern matters more than any individual typo: **the portfolio repeatedly calculates a perfectly legitimate descriptive statistic and then promotes it one rung too far up the inferential ladder.**

Band contrast becomes molecular significance; a reduction range becomes systematic uncertainty; catalogue alternatives become epochs; an injected demonstration becomes validated recovery. That is the main thing I would try to eliminate before an expert reviewer ever opens these repositories.

Highest-impact improvements

First, perform a ruthless factual/provenance audit of every headline before doing any new science. This has the highest return by far. Fix HAT-P-11b's CARMENES attribution and make the reproduced quantity match—or explicitly differ from—the published $1.08 \pm 0.05\%$ result. Rebuild 55 Cnc e from the actual individual eclipse measurements rather than NASA Archive alternative literature values. Rewrite TRAPPIST-1e around the 2025–2026 NIRSpec results. Correct the $M_{\star}^{-2/3}$ RV scaling. Update WASP-107b's methane status. These are comparatively easy corrections, and leaving any one of them in place undermines confidence in the other fourteen projects. 29

Second, reserve the word “detection significance” for an actual hypothesis/model comparison. For HD 189733b, call the current results “empirical band-contrast S/N” and separately reproduce the literature's molecule-removal retrieval/model comparison if you want molecular sigma values. For HAT-P-11b, derive a statistic that matches the published measurement definition. For transit and direct imaging simulations, distinguish raw aperture/depth S/N from false-alarm-calibrated search significance. This single terminology change would make the portfolio look much more scientifically mature.

Third, propagate the information that already exists in the input data rather than throwing it away. The WASP-121b and GJ 1214b data products include posterior/uncertainty information; use it. Monte-Carlo sample the fitted fluxes and $R_p/R_{\star}$, propagate asymmetric uncertainties through the nonlinear Planck inversion, and report posterior credible intervals on the day/night contrast. Never clamp a negative flux into a positive “measurement”; report a posterior upper limit where appropriate. For eclipse temperatures, integrate the stellar and planetary intensity over the actual instrumental throughput rather than treating an entire broad filter as a monochromatic measurement at its central wavelength. 30

Fourth, make covariance a first-class object in the atmospheric-spectrum projects. At minimum, every README should say explicitly whether the supplied points can be treated as independent. Where covariance matrices exist, use them:

$$\chi^2 = (\mathbf{d} - \mathbf{m})^T \mathbf{C}^{-1} (\mathbf{d} - \mathbf{m}).$$

For K2-18b in particular, do not obtain spectacular precision by summing thousands of native points under an independence assumption. Bin to approximately independent resolution elements or use the covariance/systematics framework of the underlying reduction. The same principle applies to limb spectra and neighbouring spectral bins.

Fifth, add an explicit “validation against the paper” block to every observational script. Each script should finish by comparing its derived quantity with the exact corresponding published value: for example, “our weighted statistic is X; paper reports Y; these are/are not the same estimator, because...”. Automated assertions can catch gross mistakes in units, wavelength definitions and data provenance. HAT-P-11b would probably never have reached the current public form had an assertion expected the paper's $1.08 \pm 0.05\%$ combined absorption. 31

Sixth, make the four detection simulations genuinely blind. This would substantially improve them without creating new repositories. For transit photometry, randomly draw the period, epoch, duration and depth, hide them from the search function, include gaps and correlated noise, scan duration as well as period/phase, and estimate false-alarm performance over many noise-only trials. For RV, impose physically meaningful parameter bounds, fit a jitter term, calculate a periodogram FAP, and run a Monte Carlo ensemble over signal amplitudes down to challenging regimes. For microlensing, remove every use of `TRUE_Q` and `TRUE_TP` from detection/recovery, and preferably inject an actual binary-lens light curve using a standard solver while keeping your PSPL fit as the null model. For imaging, inject realistic PSF-correlated and time-varying speckles, recover sources blindly, calibrate algorithm throughput via fake-planet injection/recovery, and apply the Mawet small-sample correction. 32

Seventh, upgrade one existing planet repository from “published spectrum → secondary statistic” to a genuinely end-to-end analysis. One excellent repository that starts with calibrated JWST time-series exposures, extracts a white/spectroscopic light curve, models instrumental baseline terms, fits the transit/eclipse, derives the spectrum and validates against the published reduction would be stronger evidence of research readiness than another ten template atmosphere pages. The WASP-96b or LHS 475b projects would be sensible candidates because they lead naturally to explicit null/model-comparison tests.

Eighth, strengthen software reproducibility. Replace “queried live” wherever the scientific result depends on a value that can change with NASA Archive defaults. Archive the exact TAP query result used for the calculation, record retrieval date, publication reference and units, and preferably SHA-256 hashes for downloaded Zenodo products. Pin environments rather than merely listing broad dependencies; add tests for unit conversion, Planck inversion, Kepler solutions and known limiting cases; and make the analysis fail loudly on invalid input instead of silently clipping it. A researcher should be able to rerun the repository a year later and get the same headline values.

Ninth, make the webpages less promotional and more paper-like. The visual design is clean, consistent and better than most student GitHub pages, but phrases such as “real”, “genuine”, “direct”, “caught in the act”, “decisively” and “reproduces” occur so frequently that they start to read as marketing rather than scientific writing. A stronger standard template would be: *Data product → exact estimator → assumptions → numerical result → comparison with publication → what this calculation cannot establish*. The portfolio would look more expert with fewer badges and more diagnostic plots/residuals.

Tenth, curate rather than present all fifteen as equally authoritative. Once corrected, I would foreground perhaps three observational projects and two detection-method projects whose methods are genuinely strongest, and treat the others as supporting exercises. At present, a reviewer who happens to open LHS 475b or HD 209458b first may come away favourably impressed; one who opens microlensing, HAT-P-11b or 55 Cnc e first may question the reliability of the entire portfolio. In an application, portfolio quality is set disproportionately by the weakest prominently displayed project.

Final verdict

C+ / B– as it stands: unusually ambitious and clearly based on genuine quantitative work, but not yet research-grade because several headline inferences are statistically overstated and there are a few serious provenance/factual errors; the HAT-P-11b instrument attribution, 55 Cnc e “two-epoch” analysis, TRAPPIST-1e literature claim, HD 189733b molecular sigma labels, and circular microlensing recovery are the items that would genuinely embarrass the author in front of a domain expert and should be fixed before this portfolio is used for graduate-school, RA or astronomy-facing job applications.

-
- 1 15 26 **A JWST transmission spectrum of the nearby Earth-sized exoplanet LHS 475 b | Nature Astronomy**
https://www.nature.com/articles/s41550-023-02064-z?utm_source=chatgpt.com
 - 2 28 **A warm Neptune's methane reveals core mass and vigorous atmospheric mixing | Nature**
https://www.nature.com/articles/s41586-024-07395-z?utm_source=chatgpt.com
 - 3 27 **Supersolar metallicity and tentative evidence for photochemistry on WASP-96 b from JWST and ground-based VLT transmission spectroscopy - University of St Andrews Research Portal**
https://research-portal.st-andrews.ac.uk/en/publications/supersolar-metallicity-and-tentative-evidence-for-photochemistry-/?utm_source=chatgpt.com
 - 4 (PDF) **Investigating aerosols as a way to reconcile K2-18 b JWST MIRI and NIRISS/NIRSpec observations**
https://www.researchgate.net/publication/396452995_Investigating_aerosols_as_a_way_to_reconcile_K2-18_b_JWST_MIRI_and_NIRISSNIRSpec_observations?utm_source=chatgpt.com
 - 5 24 **Magnesium Silicate Clouds in the Atmosphere of HD 209458b from a Rule-Based Tree-Structured Data Reduction**
https://arxiv.org/abs/2606.00177?utm_source=chatgpt.com
 - 6 21 **JWST TRAPPIST-1 e/b Program: Motivation and First Observations - University of Bristol**
https://research-information.bris.ac.uk/en/publications/jwst-trappist-1-eb-program-motivation-and-first-observations/?utm_source=chatgpt.com
 - 7 **SiO and a super-stellar C/O ratio in the atmosphere of the giant exoplanet WASP-121 b | Nature Astronomy**
<https://www.nature.com/articles/s41550-025-02513-x>
 - 8 30 **A reflective, metal-rich atmosphere for GJ 1214b from its JWST phase curve | Nature**
https://www.nature.com/articles/s41586-023-06159-5?utm_source=chatgpt.com
 - 9 20 **Hydrogen sulfide and metal-enriched atmosphere for a Jupiter-mass exoplanet | Nature**
https://www.nature.com/articles/s41586-024-07760-y?utm_source=chatgpt.com
 - 10 19 **Variability in the super-Earth 55 Cnc e | Monthly Notices of the Royal Astronomical Society | Oxford Academic**
https://academic.oup.com/mnras/article-abstract/455/2/2018/1106184?utm_source=chatgpt.com
 - 11 (PDF) **The Transient Outgassed Atmosphere of 55 Cancri e**
https://www.researchgate.net/publication/374612142_The_Transient_Outgassed_Atmosphere_of_55_Cancri_e?utm_source=chatgpt.com
 - 12 **Confirming Variability in the Secondary Eclipse Depth of the Super-Earth 55 Cancri e**
https://arxiv.org/abs/1804.03735?utm_source=chatgpt.com
 - 13 14 29 31 **Spectrally resolved helium absorption from the extended atmosphere of a warm Neptune-mass exoplanet**
https://arxiv.org/abs/1812.02189?utm_source=chatgpt.com
 - 16 **Periodogram Overview**
https://exoplanetarchive.ipac.caltech.edu/docs/pgram/pgram_overview.html?utm_source=chatgpt.com
 - 17 **Publications | Gemini Observatory**
https://www.gemini.edu/science/publications/publication/221?utm_source=chatgpt.com
 - 18 22 32 **Fundamental limitations of high contrast imaging set by small sample statistics**
https://arxiv.org/abs/1407.2247?utm_source=chatgpt.com

23 K2-18b Does Not Meet the Standards of Evidence for Life – DOAJ

https://doaj.org/article/35a01ea8928e4baf8e7e2b4976125dad?utm_source=chatgpt.com

25 Periodogram Algorithms

https://exoplanetarchive.ipac.caltech.edu/docs/pgram/pgram_algo.html?utm_source=chatgpt.com